

Exercise 1

Exercise 1.1 Optimal points

Consider the following optimization problem from [1, Exercise 4.1]

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 1 \\ & x_1 + 3x_2 \geq 1 \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

Draw the admissible set and determine the optimal points $\mathbf{x}^*$ and the optimal cost f^* (graphically using the contour lines) for each of the following cost functions:

$$f(\mathbf{x}) = x_1 + x_2 \tag{1}$$

$$f(\mathbf{x}) = -x_1 - x_2. \tag{2}$$

Add-On: Examine the functions:

$$f(\mathbf{x}) = x_1 \tag{3}$$

$$f(\mathbf{x}) = \max\{x_1, x_2\}. \tag{4}$$

Exercise 1.2 Convexity

Examine whether the following functions are convex or concave on their domain of definition:

$$f(\mathbf{x}) = \frac{x_1^2}{x_2}, \quad x_1, x_2 > 0 \tag{5}$$

$$f(\mathbf{x}) = \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right), \quad \mathbf{x} \in \mathbb{R}^2. \tag{6}$$

Exercise 1.3 Line search

The following unconstrained parameter optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^2} (x_1 - 1)^2 + x_1 x_2 + (2x_2 - 4)^2 \tag{7}$$

is to be solved with a line search method starting at the point $\mathbf{x}^0 = [-1, 1]^T$. Compute the first two steps of the gradient method and the Newton's method. To determine the step size, solve the underlying scalar optimization problem exactly.

Add-On: Compute the first two steps of the conjugate gradient method.

References

- [1] S. Boyd und L. Vandenberghe. *Convex Optimization*. http://www.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf. Cambridge: Cambridge University Press, 2004.